TRADECRAFT OS

The Curriculum Operating System for the Modern Trades

Forged on bare metal inside The Fortress v1.26. Built by an EPA-certified Journeyman Electrician and systems architect to carry AI from the classroom straight into the working man’s toolbox.

EPA Certified & Journeyman Electrician | **Forged on Bare Metal** | **Zero-IT Local Edge Nodes** | **Classroom to Jobsite Worldwide**

Michael Gorby Jr. • Journeyman Electrician • EPA Certified • Founder & Lead Systems Architect

ORIGIN STORY & EXECUTIVE VISION • TRADECRAFT LINEAGE & THE FORTRESS v1.26

Trades Were Life. Computers Were Passion. Bringing Them Together.

> *“Computers were my hobby since thirteen; the trades were my life. As an EPA-certified Journeyman Electrician who can build a home from the ground up, I didn’t build TradeCraft OS in a corporate boardroom—I forged it on bare metal to bridge deep tradecraft lineage with sovereign edge AI.”*

We don’t pretend to be a bloated corporate giant, and we aren't software tourists guessing what a trade school needs. As a licensed tradesman, educator, and systems architect, I built TradeCraft OS from the ground up to eliminate the outdated friction holding back adult trade education—delivering intelligent curriculum workflows on affordable, zero-IT local hardware. AI is coming to the jobsite in a profound way. We give students the infrastructure to understand it today, so they wield it as their sharpest tool tomorrow.

Master Tradesman Lineage

EPA Universal certified & Journeyman Electrician. Built homes from the ground up—structural framing, 200A service, rough-in, line sets, and finish.

Bare-Metal Homelab Origin

20+ years of hardware obsession. Self-hosted Linux and Proxmox foundation where all twelve core engines were engineered and stress-tested.

The Sovereign Convergence

Where physical trade discipline meets bare-metal edge AI. Real tools built for craftsmen at the bench, not white-collar cloud tourists.

In plain English: Built by a Journeyman Electrician who builds homes from the ground up and architects bare-metal servers—bringing trades and technology together as one.

02 / STRATEGIC FOOTPRINT & GROWTH PHILOSOPHY

Grounded Today. Built to Scale Tomorrow.

A disciplined, capital-efficient expansion model designed to maximize impact for working trade students.

STAGE 01 — 01

THE FORGE

The Fortress v1.26 & Tradecraft Lineage

Custom self-hosted bare-metal homelab where tradecraft domain mastery and bare-metal systems engineering united to prove our core engines under real loads.

STAGE 02 — 02

LOCAL HQ HUBS

Strategic Student Reach

Local operating headquarters placed directly in working trade communities where we serve the maximum concentration of trade schools, apprentices, and instructors.

STAGE 03 — 03

FUTURE HORIZON

Corporate R&D Campus

Earned expansion into a dedicated corporate R&D facility and hardware manufacturing line as worldwide campus adoption and contractor tool licensing scale.

In plain English: We anchor our operational hubs where the workers and students are—earning our growth through customer cash flow and genuine utility.

Zero Vanity Overhead

Capital Efficient

Overhauling Adult Trade Education

Transforming outdated vocational classrooms into sovereign, high-velocity technical proving grounds — built by tradesmen, for tradesmen.

✕ Legacy Vocational Schooling

Outdated Curriculums
Disconnected from modern field realities, NEC updates, and smart mechanical systems.

Fragile Cloud Technology
Requires costly on-site IT technicians; classes instantly grind to a halt during internet drops.

Passive Learning Drag
Static three-ring binders and manual grading queues steal valuable instructor time from the bench.

Zero AI Readiness
Graduates step into modern industrial facilities completely unprepared for computerized diagnostics.

✓ The TradeCraft OS Standard

Intelligent Integrations
Unified curriculum OS automatically wiring lessons, rubrics, testing, and rosters together.

Technician-Free Hardware
Affordable plug-and-play local appliances that boot up and self-configure without IT staff.

True Edge Computing
Instantaneous sub-millisecond local execution; 100% functionality with the internet severed.

Jobsite Continuum
Software that travels with the student from the lab table directly into their field toolbag.

In plain English: Every minute a vocational teacher spends fighting computer software or grading paperwork is a minute stolen from hands-on trade training.

04 / ARCHITECTURAL PILLARS

Three Engines Driving TradeCraft OS

The architectural triad bridging vocational classroom theory, zero-maintenance local hardware, and active jobsite tools.

Intelligent Curriculum OS

AI-integrated trade schooling workflows that turn raw standards into structured modules, labs, and interactive drills.

- Automatic rubric mapping & one-touch grading
- In-memory attendance & state audit compliance
- 24/7 personalized offline Study Buddy mentoring

Zero-IT Local Hardware

Affordable plug-and-play classroom nodes engineered for extreme reliability and zero administrative maintenance.

- Dual-redundant on-premise failover architecture
- Instantaneous zero-latency local LLM inference
- Zero cloud subscription bills or per-token charges

Edge-to-Toolbox Continuity

Classroom software specifically built to accompany the student into professional field work after graduation.

- Lifelong verified student competency dossier
- Real-time NEC code calculations & fault finding
- The universal diagnostic companion on the truck

In plain English: A unified software and hardware ecosystem that begins on day one of trade school and protects the craftsman across a 30-year career.

05 / ON-SITE EDGE DEPLOYMENT

Affordable, Technician-Free Infrastructure

Deploying enterprise-grade AI directly to vocational trade schools without the crushing expense of IT departments.

By strategically placing our local headquarters near working trade communities—and shipping self-contained hardware nodes—we serve the maximum number of students without bloated overhead.

0 IT

Zero IT Staff Required

Self-contained, auto-configuring appliance boots up and serves the classroom in under 60 seconds without network reconfiguration.

100%

Local Sovereign Execution

Zero cloud dependency, zero per-query token fees, zero latency lag, and complete immunity to outside internet outages.

v1.26

Fortress Engineered Core

Distilled directly from our battle-tested bare-metal architecture into ruggedized, low-power edge compute units.

Trade schools do not need a five-person IT support staff. They need an appliance that powers up and **just works every single day**.

Into the Working Man's Everyday Toolbox

> *"Our vision doesn't end at graduation. As someone who has worked the trades and built homes from the ground up, I know real trade mastery is proven in the field. TradeCraft travels with the apprentice straight into their everyday toolbag."*

The same system that guides an apprentice through their very first conduit bend becomes the diagnostic companion they trust on the commercial jobsite for the rest of their career.

STEP 01 // LAB MASTERY

Learn in the Lab

Master trade theory, safety protocols, and NEC standards on TradeCraft OS with instantaneous AI coaching and hands-on shop simulations.

STEP 02 // CREDENTIAL TRANSFER

Graduate with the OS

Carry your verified digital competency dossier, shop lab portfolio, and state licensing hours directly into your professional field device.

STEP 03 // JOBSITE DEPLOYMENT

Wield in the Toolbox

Solve live electrical faults, calculate 3-phase motor loads, verify clearance codes, and order parts directly from the service truck.

In plain English: Vocational schools are our customer acquisition engine; working contractors are our lifelong enterprise community.

07 / R&D IN DEVELOPMENT: SPATIAL LAB ASSISTANTS

AI That Sees the Lab & Tutors the Student

Moving beyond text prompts: spatially aware computer vision that turns a student's phone photo into live physical tutoring.

LIVE LAB ASSISTANT PIPELINE // "Why Does It Go Here?"

Students don't just get a sterile text answer—the spatial engine overlays 3D physical routing vectors, NEC code logic, and spatial reasoning over their actual workspace while instructors focus where master craftsmanship matters most.

Photo-to-Spec Drawings

Instant schematics and wiring diagrams overlaid directly onto physical conduit boxes and subpanels.

Live NEC Code Engine

Real-time box-fill calculations, wire ampacity checks, and automated code compliance references.

Physical Vector Mapping

Visual 3D bend vectors, take-up calculations, and obstacle routing mapped onto conduit runs.

Instructor Multiplier

AI handles routine mechanical verification and math questions so teachers coach high-level tradecraft.

In plain English: A student points their camera at a bent pipe or wire run, and the AI immediately explains the spatial geometry and NEC code requirements.

08 / MULTI-MODAL FIELD INTELLIGENCE

Strapping Eyes & Sensors Onto Jobsite AI

Equipping next-generation rugged edge hardware with industrial sensors to diagnose physical world problems in real time.

3D LiDAR & Point Clouds

Millimeter-accurate spatial room scanning for conduit routing, pipe runs, and structural clearances before cutting a single piece of stock.

Thermal Heat-Mapping

Real-time electrical load hotspot detection, unbalanced 3-phase distribution, and failing circuit breaker diagnostics under live conditions.

Ultrasonic & Acoustic

Pinpointing high-voltage arc-flash signatures, pressurized refrigerant line leaks, and bearing friction wear before catastrophic failure occurs.

Live CAD Photogrammetry

Instant bend angle verification, 3D clearance calculations, and automated as-built compliance drawings generated straight from jobsite video.

Industrial sensor fusion gives the AI **true physical senses**—diagnosing invisible jobsite hazards before human tools ever touch live wire.

From Local Student Hubs to Tool Shelves Worldwide

A disciplined, three-stage roadmap translating grounded trade education into worldwide hardware deployment.

With your investment, we scale responsibly—placing our local headquarters where it serves the most students today, while building the flagship technology sold in tool stores worldwide tomorrow.

GROWTH STAGE	TECHNOLOGY DELIVERABLE	FOOTPRINT & DISTRIBUTION CHANNEL
1 **Stage 1: Grounded Rollout**	**TradeCraft OS + Zero-IT Local Hardware** 12-engine curriculum monolith, DocuAI scan-to-grade, and on-premise classroom nodes.	**Strategic Local HQ Hubs** Direct deployment into regional trade campuses, vocational career centers, and adult trade schools.
2 **Stage 2: Industry Standard**	**Spatially Aware Lab Assistant (NEC + Vectors)** Computer vision workbench tutoring, automated code verification, and mobile companion app.	**Regional Training Hubs & Unions** National apprenticeship programs, trade union locals (IBEW, UA, SMART), and licensing bodies.
3 **Stage 3: Global Scale**	**Multi-Sensor Field AI (LiDAR / Thermal / Rugged Hardware)** Industrial diagnostic handhelds, predictive fault telemetry, and contractor fleet management.	**Corporate Campus & Retail Tool Shelves** Flagship R&D manufacturing campus; global distribution via Home Depot, Grainger, and Klein Tools.

In plain English: A clear path from captive classroom revenue to multi-billion dollar industrial tool distribution.

TradeCraft OS

> *“Computers were my hobby. The trades were my life. As an EPA-certified Journeyman Electrician who can build a home from the ground up, I forged TradeCraft OS to bring those two worlds together—equipping working tradesmen with the sharpest AI tools on Earth.”*

1
Master Craftsmanship
Authentic tradecraft lineage: EPA certified, Journeyman Electrician, ground-up home builder.

2
Sovereign Moat
Zero cloud token burn, bare-metal local hardware, and deeply captive trade curriculum datasets.

3
Worldwide Horizon
Carrying AI from local vocational classrooms straight into millions of working toolboxes worldwide.

TradeCraft OS // Forged in The Fortress v1.26 // Built for the World’s Toolboxes

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